ДЗ №2

№ 1837

$$\int x \operatorname{arctg} x\,dx = \frac{1}{2}\int \operatorname{arctg} x\,dx^2 = \frac{1}{2}\left(\operatorname{arctg} x \cdot x^2 - \int x^2\,d\operatorname{arctg} x\right) =$$

$$= \frac{1}{2}\operatorname{arctg} x \cdot x^2 - \frac{1}{2}\int \frac{x^2\,dx}{x^2+1} = \frac{1}{2}\operatorname{arctg} x\, x^2 - \frac{1}{2}x + \frac{1}{2}\int\frac{dx}{1+x^2} =$$

$$= \frac{1}{2}\operatorname{arctg} x\, x^2 - \frac{1}{2}x + \frac{1}{2}\operatorname{arctg} x + C$$

№ 1838

$$\int \arccos x\,dx = \arccos x \cdot x - \int x\,d\arccos x = x\cdot\arccos x -$$

$$-\int x\cdot\left(-\frac{1}{\sqrt{1-x^2}}\,dx\right) = x\arccos x + \frac{1}{2}\int\frac{dx^2}{\sqrt{1-x^2}} = x\arccos x +$$

$$+\frac{1}{2}\int\frac{-1\cdot d(1-x^2)}{(1-x^2)^{1/2}} = x\arccos x - \frac{1}{2}\,\frac{(1-x^2)^{1/2}}{1/2} = x\arccos x -$$

$$-\sqrt{1-x^2} + C$$

№ 1839

$$\int \operatorname{arctg}\sqrt{x}\,dx = \left[\begin{array}{l} t^2 = x \\ dx = dt^2 = 2t\,dt \end{array}\right] = \int \operatorname{arctg} t \cdot 2t\,dt =$$

$$= \frac{2}{2}\int \operatorname{arctg} t\,dt^2 = \int \operatorname{arctg} t\,dt^2 = \operatorname{arctg} t \cdot t^2 - \int t^2\,d\operatorname{arctg} t =$$

$$= t^2\operatorname{arctg} t - \int\frac{t^2}{1+t^2}\,dt = t^2\operatorname{arctg} t - \int\left(1 - \frac{1}{1+t^2}\right)dt =$$

$$= t^2\operatorname{arctg} t - t + \int\frac{dt}{1+t^2} = t^2\operatorname{arctg} t - t + \operatorname{arctg} t + C =$$

$$= \operatorname{arctg} t\cdot(t^2+1) - t + C = \operatorname{arctg}(\sqrt{x})\cdot(x+1) - \sqrt{x} + C$$

№ 1850

$$\int x^2 e^{-x}\,dx = \int x^2\cdot(-1)\,d(e^{-x}) = -\int x^2\,de^{-x} = -x^2 e^{-x} + \int e^{-x}\,dx^2 =$$

$$= 2\int e^{-x}\cdot x\,dx - x^2 e^{-x} = -2\int x\,de^{-x} - x^2\cdot e^{-x} = -2xe^{-x} + \int e^{-x}\,dx -$$

$$- x^2 e^{-x} = -2e^{-x}x - 2e^{-x} - x^2 e^{-x} + C = -e^{-x}(2 + 2x + x^2) + C$$

1861

$$\int e^{3x}(\sin 2x - \cos 2x)\,dx = \overbrace{\int e^{3x}\sin 2x\,dx}^{I} - \overbrace{\int e^{3x}\cos 2x\,dx}^{II} =$$

$$= -\frac{1}{2}\int e^{3x}\,d\cos 2x - \frac{1}{2}\int e^{3x}\,d\sin 2x = \frac{1}{2}\left(e^{3x}\cos 2x - \int \cos 2x\,de^{3x}\right) -$$

$$-\frac{1}{2}\left(e^{3x}\sin 2x - \frac{3}{2}\int \sin 2x\,de^{3x}\right) = -\frac{1}{2}\left(e^{3x}\cos 2x - \frac{3}{2}\int e^{3x}\,d\sin 2x\right)$$

$$-\frac{1}{2}\left(e^{3x}\sin 2x + \frac{3}{4}\int e^{3x}\,d\cos 2x\right) = -\frac{1}{2}\Big(e^{3x}\cos 2x -$$

$$-\frac{3}{2}\left(e^{3x}\sin 2x - \int \sin 2x\,de^{3x}\right)\Big) - \frac{1}{2}\Big(e^{3x}\sin 2x + \frac{3}{2}e^{3x}\cos 2x -$$

$$-\frac{3}{2}\cdot 3\int e^{3x}\cos x\,dx\Big) \Rightarrow$$

$$I + \frac{9}{4}I = \frac{1}{2}e^{3x}\cos 2x + \frac{3}{4}e^{3x}\sin 2x$$

$$I = \frac{4}{13}\cdot\left(-\frac{1}{2}e^{3x}\cos 2x + \frac{3}{4}e^{3x}\sin 2x\right)$$

$$I = \frac{e^{3x}}{13}(3\sin 2x - 2\cos 2x) + C$$

$$II + \frac{9}{4}II = \frac{1}{2}e^{3x}\sin 2x + \frac{3}{4}e^{3x}\cos 2x$$

$$II = \frac{e^{3x}}{13}(2\sin 2x + 3\cos 2x)$$

$$I - II = \frac{e^{3x}}{13}(\sin 2x - 5\cos 2x) + C$$

1871

$$\int \frac{4x+3}{(x-2)^3}\,dx = \begin{bmatrix} x-2=t \to x=t+2 \\ dx = d(t+2) = dt \end{bmatrix} = \int \frac{4t+11}{t^3}\,dt =$$

$$= 4\int \frac{4}{t^2}\,dt + \int \frac{11}{t^3}\,dt = \frac{4\cdot t^{-2+1}}{-1} + \frac{11\cdot t^{-3+1}}{-2} + C = -\frac{4}{t} - \frac{11}{2t^2} =$$

$$= \frac{-4}{x-2} - \frac{11}{2(x^2-4x+4)} = \frac{-4}{x-2} - \frac{11}{2x^2-8x+8} + C$$

№ 1869

$$\int \frac{dx}{1+\sqrt{x+1}} = \begin{bmatrix} z^2 = x+1 \\ dx = d(z^2-1) = 2z\,dz \end{bmatrix} = \int \frac{2z\,dz}{1+z} = 2\int dz -$$

$$-2\int \frac{dz}{1+z} = 2z - 2\ln|1+z| = 2\sqrt{x+1} - 2\ln|1+\sqrt{x+1}|$$

1875

$$\int \frac{\sqrt{x}}{x(x+1)}dx = \begin{bmatrix} x = z^2 \\ dx = 2z\,dz \end{bmatrix} = \int \frac{z \cdot 2z\,dz}{z^2(z^2+1)} = 2\int \frac{dz}{z^2+1} =$$

$$= 2\,\mathrm{arctg}\, z = 2\,\mathrm{arctg}\sqrt{x} + C$$

~~1876~~ №1876

$$\int \frac{\sqrt{x}}{x+1}dx = \begin{bmatrix} z^2 = x \\ dx = 2z\,dz \end{bmatrix} = \int \frac{z \cdot 2z\,dz}{z^2+1} = \int 2dz - \int \frac{2dz}{z^2+1} =$$

$$= 2z - 2\,\mathrm{arctg}\, z + C = 2\sqrt{x} - 2\,\mathrm{arctg}\sqrt{x} + C$$

1884

$$\int \frac{dx}{\sqrt{1+e^x}} = \begin{bmatrix} 1+e^x = z^2 \\ dz = d((1+e^x)^{\frac{1}{2}}) = \frac{1}{2}e^x \frac{dx}{\sqrt{1+e^x}} \end{bmatrix} = \int \frac{2dz}{z^2-1} = \frac{2}{2}\ln\left|\frac{z-1}{z+1}\right| =$$

$$= \ln\left|\frac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1}\right| + C$$

№1909

$$\int \frac{\mathrm{arctg}\, x}{x^2(1+x^2)}dx = \begin{bmatrix} \mathrm{arctg}\, x = t & \mathrm{tg}\, t = x \\ dt = \frac{dx}{1+x^2} \end{bmatrix} = \int \frac{t\,dt}{\mathrm{tg}^2 t} = \int \mathrm{ctg}^2 t \cdot t\,dt =$$

$$= \frac{1}{2}\int \mathrm{ctg}^2 t\,dt^2 = \frac{1}{2}\mathrm{ctg}^2 t\, t^2 - \frac{1}{2}\int t^2\,d\,\mathrm{ctg}^2 t = \int \frac{t}{\sin^3 t}dt -$$

$$-\int t\,dt = t\,\mathrm{tg}\, t + \ln|\cos t| = \mathrm{arctg}\, x \cdot x + \ln|\cos(\mathrm{arctg}\, x)| + C$$

№2012

$$\int \frac{x\,dx}{(x+1)(2x+1)} =$$

$$\frac{x}{(x+1)(2x+1)} = \frac{A}{x+1} + \frac{B}{2x+1} \Rightarrow 2Ax + A + Bx + B = x$$

$$\begin{cases} 2A + B = 1 \\ A + B = 0 \end{cases} \Rightarrow \begin{cases} A = -B \\ A = 1 \\ B = -1 \end{cases} \Rightarrow = \int\left(\frac{1}{x+1} + \frac{-1}{2x+1}\right)dx =$$

$$= \ln|x+1| - \frac{1}{2}\ln|2x+1| + C$$

2013

$$\int \frac{x\,dx}{2x^2-3x-2} = \int \frac{x\,dx}{2(x-2)(x+\frac{1}{2})} = \frac{1}{2}\int\left(\frac{A}{x-2}+\frac{B}{x+\frac{1}{2}}\right)dx =$$

$$\left(x+\frac{1}{2}\right)A + B(x-2) = x \Rightarrow \begin{cases} A+B=1 \\ \frac{1}{2}A-2B=0 \end{cases} \Rightarrow \begin{cases} B=\frac{1}{5} \\ A=\frac{4}{5} \end{cases}$$

$$= \frac{2}{5}\int\frac{1}{x-2}\,dx + \frac{1}{10}\int\frac{1}{x+\frac{1}{2}}\,dx = \frac{2}{5}\ln|x-2| + \frac{1}{10}\ln\left|x+\frac{1}{2}\right| + C$$

№2036

$$\int\frac{dx}{(x^2+1)x} = \int\left(\frac{A}{x}+\frac{Bx+C}{x^2+1}\right)dx =$$

$$Ax^2+A+Bx^2+Cx=1 \Rightarrow \begin{cases} A+B=0 \\ C=0 \\ A=1 \end{cases} \Rightarrow \begin{cases} A=1 \\ B=-1 \\ C=0 \end{cases}$$

$$= \int\left(\frac{1}{x}+\frac{-x}{x^2+1}\right)dx = \int\frac{dx}{x} - \int\frac{x\,dx}{x^2+1} = \ln|x| - \frac{1}{2}\ln|x^2+1| + C$$

2043

$$\int\frac{dx}{(x+1)^2(x^2+1)} = \int\left(\frac{A}{x+1}+\frac{B}{(x+1)^2}+\frac{Cx+D}{x^2+1}\right)dx =$$

$$A(x+1)^2(x^2+1) + B(x+1)(x^2+1) + (Cx+D)(x+1)^3 = 1$$

$$\begin{cases} Ax^4+Cx^4=0 \\ A2x^3+Bx^3+2Cx^3+Dx^3=0 \\ 2Ax^2+Bx^2+3Cx^2+2Dx^2=0 \\ 2Ax+Bx+Cx+3Dx=0 \\ A+B+D=1 \end{cases} \Rightarrow \begin{cases} A=1 \\ B=-D=\frac{1}{2} \\ C=-1 \\ D=-\frac{1}{2} \end{cases}$$

$$= \int\left(\frac{dx}{x+1}\right) + \frac{1}{2}\int\frac{dx}{(x+1)^2} + \int\frac{-x-\frac{1}{2}}{x^2+1}\,dx = \ln|x+1| +$$

$$+\frac{1}{2}\frac{(x+1)^{-1}}{(-1)} - \int\frac{x+\frac{1}{2}}{x^2+1}\,dx = \ln|x+1| - \frac{1}{2(x+1)} - \int\frac{x\,dx}{x^2+1} - \frac{1}{2}\int\frac{dx}{x^2+1} =$$

$= \ln|x+1| - \frac{1}{2(x+1)} - \frac{1}{2}\ln|x^2+1| - \frac{1}{2}\operatorname{arctg} x + C$

2022

$\int \frac{x^3-3x+2}{x(x^2+2x+1)}dx = \int\left(\frac{A}{x}+\frac{B}{x+1}+\frac{C}{(x+1)^2}\right) = (2)$

$\frac{A}{x}+\frac{B}{x+1}+\frac{C}{(x+1)^2} = \frac{x^3+3x+2}{x(x+1)^2}$

$A(x+1)^2 + B(x+1)x + Cx \doteq x^2-3x+2$

$\begin{cases} Ax^2+Bx^2=x^2 \\ 2xA+Bx+Cx=-3 \\ A=2 \end{cases} \Rightarrow \begin{cases} A=2 \\ B=-1 \\ C=-6 \end{cases}$

$(2)\ \int\left(\frac{2}{x}-\frac{1}{x+1}-\frac{6}{(x+1)^2}\right)dx = 2\ln|x|-\ln|x+1|-6\cdot\frac{(x+1)^{-2+1}}{(-1)}+C=$

$= 2\ln|x|-\ln|x+1|+\frac{6}{x+1}+C$